Status of COVID-19 cases in Ontario by Public Health Unit (PHU)

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```r
# loading necessary libraries
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.0     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(dplyr)
library(tidyr)
library(ggplot2)
library(ggmap)
```

```
## i Google's Terms of Service: <https://mapsplatform.google.com>
##   Stadia Maps' Terms of Service: <https://stadiamaps.com/terms-of-service/>
##   OpenStreetMap's Tile Usage Policy: <https://operations.osmfoundation.org/policies/tiles/>
## i Please cite ggmap if you use it! Use `citation("ggmap")` for details.
```

```r
library(knitr)
library(lubridate)
library(rpart)
library(randomForest)
```

```
## randomForest 4.7-1.1
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```r
library(jpeg)
```

```r
# loading data
data <- read_csv("cases_by_status_and_phu.csv")
```

```
## Rows: 33082 Columns: 8
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): PHU_NAME
## dbl  (6): PHU_NUM, ACTIVE_CASES, RESOLVED_CASES, DEATHS, ARCHIVED_RESOLVED_C...
## date (1): FILE_DATE
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# convert 'File_date' from character to Date format in "year-month-day" format
data$FILE_DATE <- as.Date(data$FILE_DATE, format = "%Y-%m-%d")

# filter data for dates within range 2022-03-11 to 2022-12-01
new_data <- data %>%
  filter(FILE_DATE >= as.Date("2022-03-11") & FILE_DATE <= as.Date("2022-12-01")) %>%
  select(!contains("ARCHIVED")) %>%  # select all variables except 'archived_resolved_cases' and 'archi
  filter(PHU_NUM %in% c(2238, 2242, 2243,2263))  # select only PHU 1, 2, 3, and 4
```

Introduction

This report is based on a dataset Status of COVID-19 cases in Ontario by Public Health Unit (PHU) sourced from Ontario Data Catalogue. The data is collected by Public Health Units (PHUs) in Ontario and The Ontario Ministry of Health with the aim of tracking COVID-19 cases across different public health units in Ontario. The dataset provides information on the status of COVID-19 cases in Ontario, including the number of active cases, resolved cases, and deaths, broken down by Public Health Unit. It serves as a valuable resource for monitoring the spread of COVID-19 and assessing the effectiveness of public health interventions and strategies. The dataset covers a specific time period with daily snapshots of COVID-19 testing and cases reported across various Public Health Units in Ontario. It includes both current and archived data on COVID-19 cases.

However, only data recorded after March 11, 2022, are included in the analysis of this report. This decision was made due to the change of the methodology used to count COVID-19 deaths by excluding deaths not caused by COVID-19, thereby ensuring a more accurate representation of the impact of the virus. As a result of the methodological change, the columns "Resolved_Cases" and "Deaths" represent data recorded after March 11, 2022, while the columns "Archived_Resolved_Cases" and "Archived_Deaths" represent data posted publicly prior to the methodological change.

```r
variables <- c("File_date", "PHU_Name", "PHU_Num", "Active_Cases", "Resolved_Cases",
               "Deaths")

types <- sapply(new_data, class)
descriptions <- c("The date of the reported data",
                  "The name of the Public Health Unit (PHU) in Ontario",
                  "A unique identifier for each Public Health Unit",
                  "Number of active COVID-19 cases",
                  "Number of resolved COVID-19 cases after March 11, 2022 (excluding deaths not caused b
                  "Number of deaths due to COVID-19 after March 11, 2022")
summary <- data.frame(Variable = variables, Types = types, Description = descriptions)
kable(summary, row.names = FALSE, caption="Variable Description")
```

Table 1: Variable Description

Variable	Types	Description
File_date	Date	The date of the reported data
PHU_Name	character	The name of the Public Health Unit (PHU) in Ontario
PHU_Num	numeric	A unique identifier for each Public Health Unit
Active_Cases	numeric	Number of active COVID-19 cases
Resolved_Cases	numeric	Number of resolved COVID-19 cases after March 11, 2022 (excluding deaths not caused by COVID)
Deaths	numeric	Number of deaths due to COVID-19 after March 11, 2022

Purpose

This report aims to answer the question, **How have the COVID-19 testing and case trends evolved across Public Health Units (PHUs) in Ontario since March 11, 2022, and what factors are associated with variations in the number of active cases, resolved cases, and deaths?**

Background

The Status of COVID-19 cases in Ontario by Public Health Unit (PHU) dataset offers critical insights into the state of the pandemic within the region. As COVID-19 continues to pose significant challenges to public health systems worldwide, understanding the distribution and trends of cases across different Public Health Units (PHUs) in Ontario is paramount for effective response and intervention strategies.

The dataset provides comprehensive information on key COVID-19 metrics, including the number of active cases, resolved cases, and deaths, reported by PHUs in Ontario. By analyzing these data points, we can gain valuable insights into the dynamics of COVID-19 transmission and its impact on various regions within the province.

To gain a deeper understanding of COVID-19 trends across PHUs in Ontario, we begin by examining summary statistics for key variables within the dataset. Table 1 presents an overview of the central tendencies and variability of COVID-19 metrics, including mean, median, minimum, and maximum values for active cases, resolved cases, and deaths recorded within the region.

```r
# summary statistics
summary_table <- data.frame(
  Variable = c("Active_cases", "Resolved_cases", "Deaths"),
  Mean = c(mean(new_data$ACTIVE_CASES), mean(new_data$RESOLVED_CASES), mean(new_data$DEATHS)),
  Median = c(median(new_data$ACTIVE_CASES), median(new_data$RESOLVED_CASES), median(new_data$DEATHS)),
  Standard_Deviation = c(sd(new_data$ACTIVE_CASES), sd(new_data$RESOLVED_CASES), sd(new_data$DEATHS)),
  Min = c(min(new_data$ACTIVE_CASES), min(new_data$RESOLVED_CASES), min(new_data$DEATHS)),
  Max = c(max(new_data$ACTIVE_CASES), max(new_data$RESOLVED_CASES), max(new_data$DEATHS))
)

# Generate the table
kable(summary_table, row.names = TRUE, caption = "Summary statistics for active cases, resolved cases, a
```

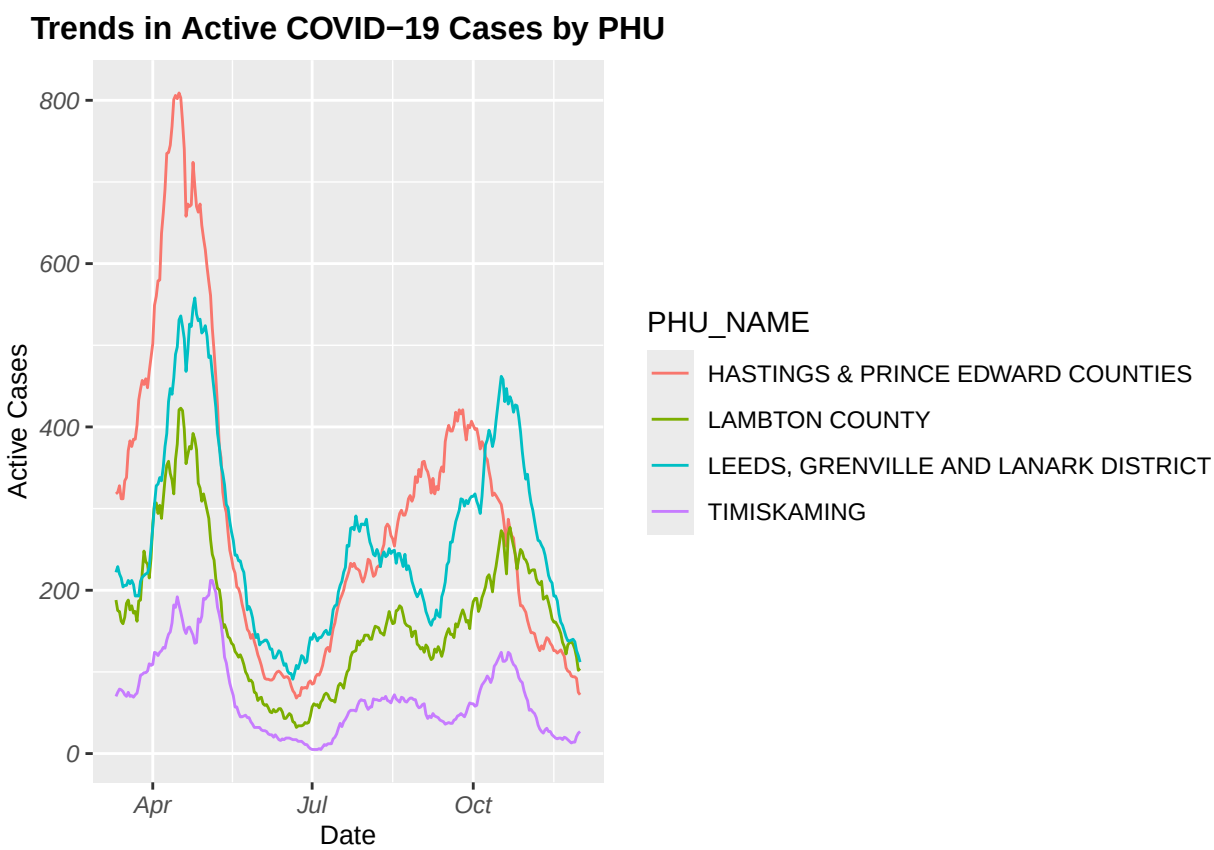
Table 2: Summary statistics for active cases, resolved cases, and deaths.

	Variable	Mean	Median	Standard_Deviation	Min	Max
1	Active_cases	198.57765	161.0	149.67621	5	809
2	Resolved_cases	8554.20455	10100.5	4106.39992	1131	13903

	Variable	Mean	Median	Standard_Deviation	Min	Max
3	Deaths	86.63163	92.0	47.79809	13	163

Next, we turn our attention to visualizing the trends in COVID-19 cases across PHUs in Ontario. Figure 1 illustrates the trajectory of active COVID-19 cases over time. This line plot captures the ebb and flow of case counts, providing insights into the temporal variations experienced by different regions.

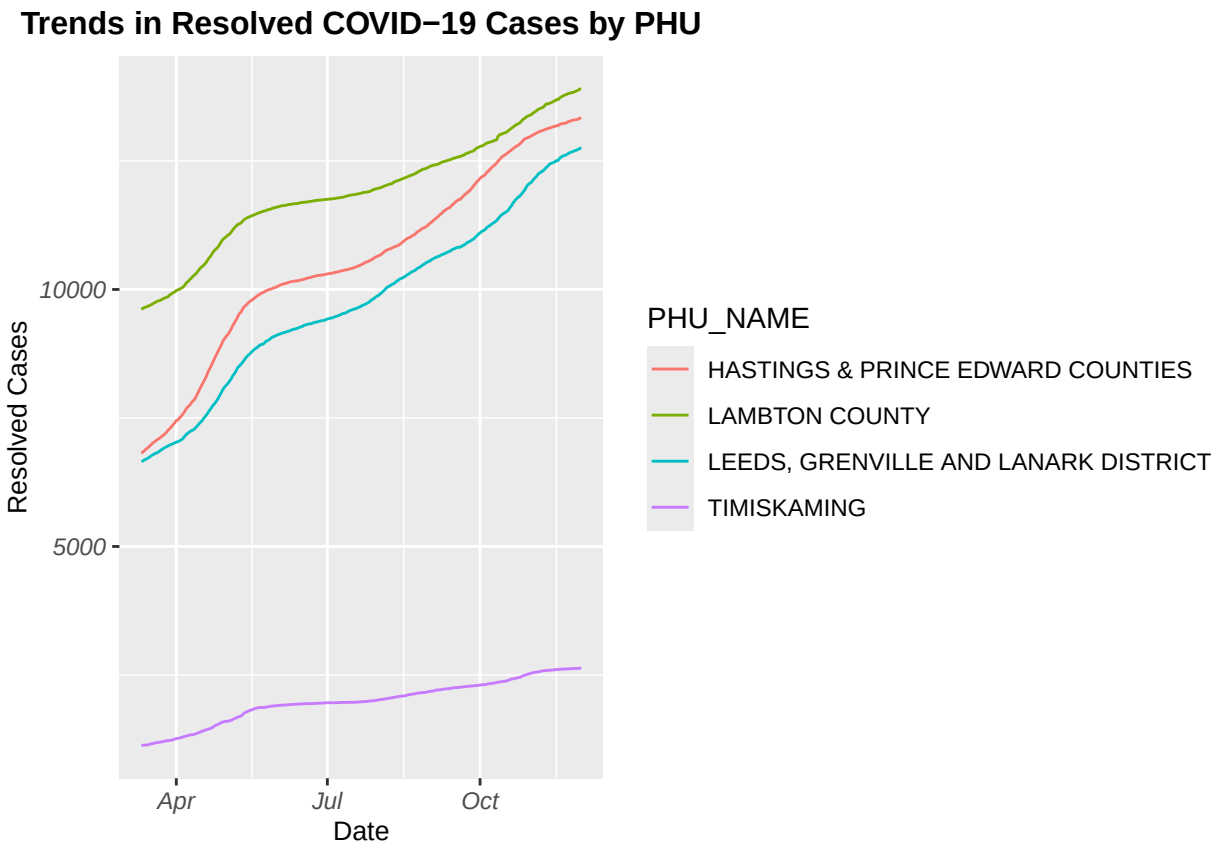
```
# create line plot for active cases by PHU
ggplot(new_data, aes(x = FILE_DATE, y = ACTIVE_CASES, group = PHU_NAME, color = PHU_NAME)) +
  geom_line() +
  labs(title = "Trends in Active COVID-19 Cases by PHU",
       x = "Date",
       y = "Active Cases") +
  theme(plot.title = element_text(face = "bold",
                                  size = 12,
                                  hjust = 0.5) ,
        axis.title = element_text(size = 10),
        axis.text = element_text(face = "italic"))
```



The fluctuating pattern observed in Figure 1 underscores the dynamic nature of COVID-19 transmission across PHUs in Ontario. Peaks and valleys in case counts suggest varying levels of contagion and highlight areas of potential concern for public health authorities.

Continuing our analysis, Figure 2 showcases the trends in resolved COVID-19 cases over time. This visualization reveals a general upward trajectory, indicating a progressive resolution of COVID-19 infections across PHUs in Ontario.

```r
# create line plot for resolved cases by PHU
ggplot(new_data, aes(x = FILE_DATE, y = RESOLVED_CASES, group = PHU_NAME, color = PHU_NAME)) +
  geom_line() +
  labs(title = "Trends in Resolved COVID-19 Cases by PHU",
       x = "Date",
       y = "Resolved Cases") +
  theme(plot.title = element_text(face = "bold",
                                  size = 12,
                                  hjust = 0.5) ,
        axis.title = element_text(size = 10),
        axis.text = element_text(face = "italic"))
```

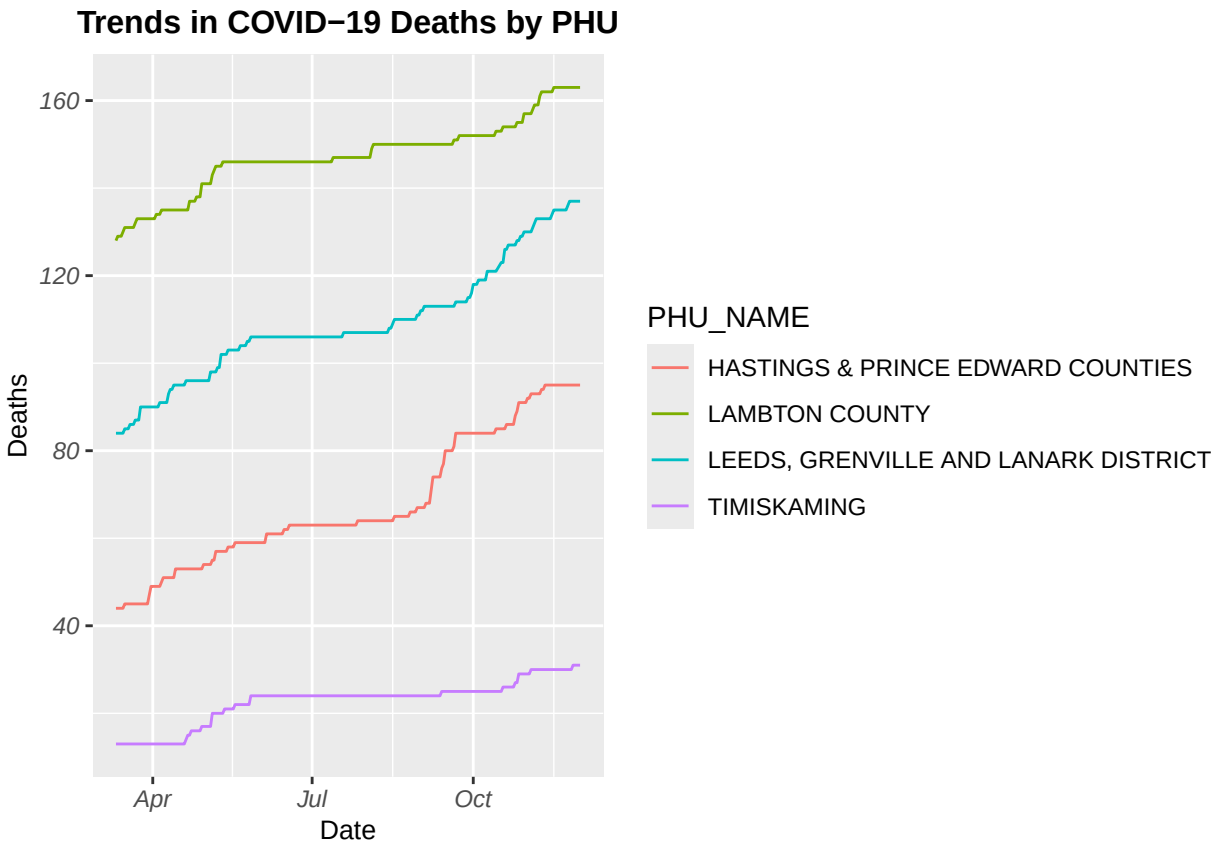


The increasing trend observed in Figure 2 reflects the collective efforts of public health initiatives and medical interventions in managing and containing the spread of COVID-19 within the region.

Lastly, Figure 3 outlines the trends in COVID-19 deaths recorded across PHUs in Ontario. While fluctuations may occur, the overall stability in mortality rates underscores the consistent impact of the virus across different regions.

```r
# create line plot for deaths by PHU
ggplot(new_data, aes(x = FILE_DATE, y = DEATHS, group = PHU_NAME, color = PHU_NAME)) +
  geom_line() +
  labs(title = "Trends in COVID-19 Deaths by PHU",
       x = "Date",
       y = "Deaths") +
  theme(plot.title = element_text(face = "bold",
                                  size = 12,
                                  hjust = 0.5) ,
```

```
        axis.title = element_text(size = 10),
        axis.text = element_text(face = "italic"))
```



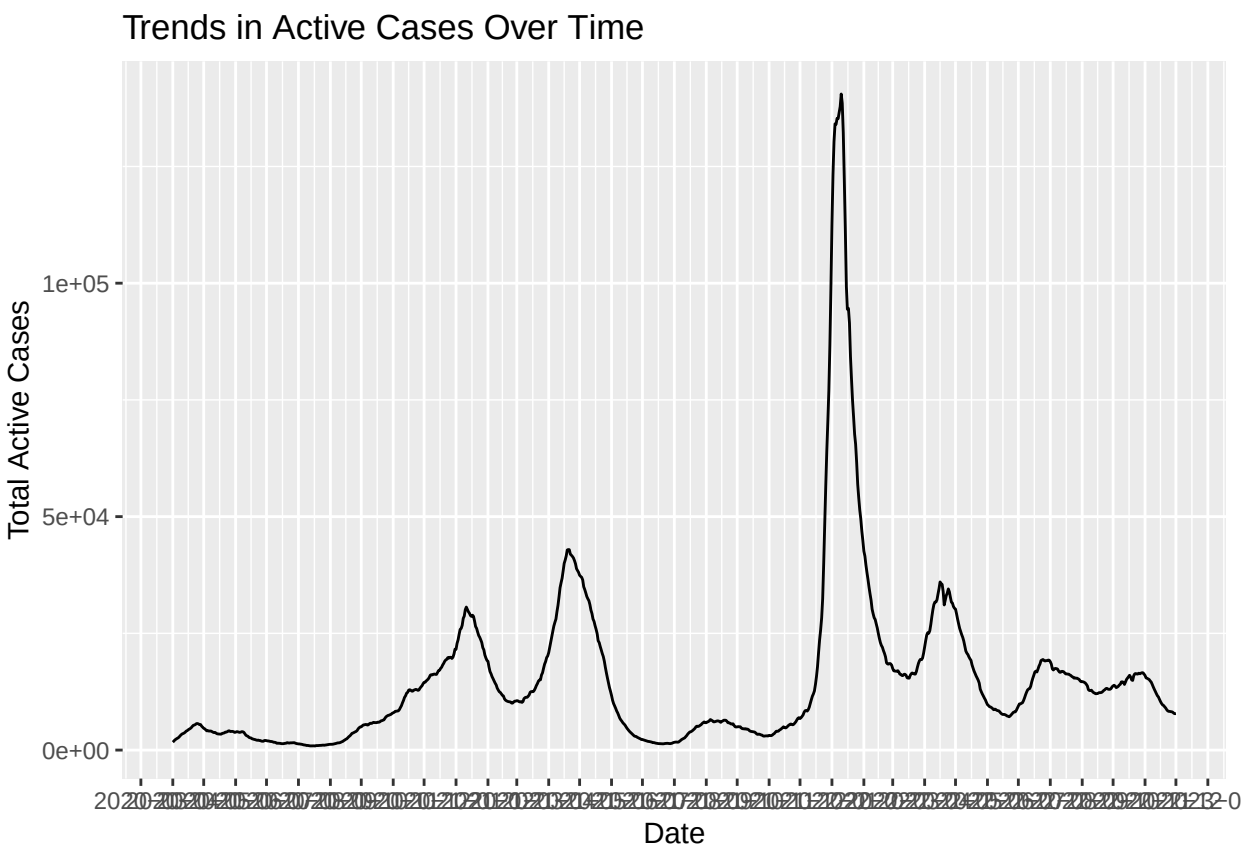
The line plot demonstrates the trends in COVID-19 deaths across PHUs in Ontario. While the number of deaths may vary between regions, we observe a relatively stable trend over time, indicating consistent mortality patterns across different areas.

These visualizations offer a nuanced understanding of the COVID-19 landscape in Ontario, providing valuable insights into the evolving trends and patterns observed across PHUs. By analyzing these trends, we can identify areas of concern and assess the effectiveness of public health interventions in mitigating the spread of the virus within the region.

```
#summary by clinic
phu_data = new_data %>% group_by(PHU_NAME) %>% summarise(TOTAL_RESOLVED=sum(RESOLVED_CASES), TOTAL_DEATH
phu_data
```

```
## # A tibble: 4 x 3
##   PHU_NAME                              TOTAL_RESOLVED TOTAL_DEATH
##   <chr>                                          <dbl>       <dbl>
## 1 HASTINGS & PRINCE EDWARD COUNTIES            2787036       17838
## 2 LAMBTON COUNTY                               3145788       38870
## 3 LEEDS, GRENVILLE AND LANARK DISTRICT         2576486       28794
## 4 TIMISKAMING                                   523930        5981
```

Graphs

```
# Aggregate data by month
data_monthly <- data %>% group_by(Month = format(FILE_DATE)) %>% summarise(Total_Active_Cases = sum(ACT
```

```r
# Plot the aggregated data
plot <- ggplot(data_monthly, aes(x = as.Date(Month), y = Total_Active_Cases)) +
          geom_line() +
          labs(title = "Trends in Active Cases Over Time",
               x = "Date",
               y = "Total Active Cases") +
          scale_x_date(date_labels = "%Y-%m", date_breaks = "1 month")

print(plot)
```